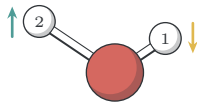


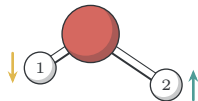
X

$$\xrightarrow[\chi_{\text{stat}}(\sigma) = -1]{\sigma = (12)}$$



$\sigma X = X P_\sigma$

$$\xrightarrow[\chi_{\pm}(E^*) = \pm 1]{E^*}$$



$-X P_\sigma = E^* \sigma X$

$$\Psi \begin{bmatrix} \uparrow & \downarrow \\ 1 & 2 \end{bmatrix} \quad \Psi(X)$$



$$-\begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix}$$

$$\Psi(X P_\sigma) = \chi_{\text{stat}}(\sigma) P_\sigma \Psi(X)$$



$$\mp \begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix}$$

$$\Psi(-X P_\sigma) = \chi_{\pm} \Psi(X P_\sigma)$$